DIGI Node — Tier 4: Bare PCB

In the box: bare AntiHunter DIGI Node PCB.

Source the parts

The full bill of materials, with part links and photos, is in **github.com/lukeswitz/AntiHunter → `hw` folder → `BOM-Links.md`**. The core parts are:

Function	Part	Function	Part
MCU	XIAO ESP32-S3	Vibration sensor	SW-420
LoRa radio	Heltec V3.2 or T114	Storage	Micro SD reader, SD card under 32 GB formatted FAT32
GPS	ATGM336H	Thermal switch	KSD9700, 40 °C
Real-time clock	DS3231	External 5 V input	Waterproof Type-C female chassis jack

Build it

1. Follow the assembly manual from start to finish: **github.com/lukeswitz/AntiHunter → `hw` folder → `Antihunter-DIGINODE-AssemblyManual.pdf`**
2. Two modules need their headers modified first (manual, page 6):
   - The RTC comes with female headers.
   - Some tamper sensors come with inverted 90° headers.
3. Fit the SD card before you close the case. You can't reach the slot once the PCB is mounted.

Flash it

4. Flash the Heltec V3.2 or T114 with the latest stable Meshtastic from flasher.meshtastic.org, then set up the serial module as below. Without these settings the node and the radio cannot communicate. `scripts/meshtastic_config.py` in the repo applies them for you.

Setting	Heltec V3	T114
Serial module	Enabled	Enabled
Mode	TEXTMSG	TEXTMSG
Baud	115200	115200
RX / TX pins	19 / 20	10 / 9

5. Flash AntiHunter on the ESP32 over USB with the web flasher at lukeswitz.github.io/AntiHunter. Choose **Full** for the web UI or **Headless** for quiet field operation.
6. Set the LoRa region and change the Bluetooth pairing PIN. Make your own encrypted channel the primary channel and turn the public default off.
7. **Full firmware only:** join the `Antihunter` Wi-Fi network with the password `antihunt3r123`, open `http://192.168.4.1` and change these credentials first.
8. Write down the erase key. The node makes one on first boot. Press **Show my erase key** in the web flasher, or read `ERASE KEY` in the serial log. Change it in the web UI under System → Settings → **Danger zone**, or over mesh with `CONFIG_ERASE_PSK`.

Power

- Use a regulated 5 V supply only: USB, a bench supply set to 5 V, a proper DC-DC regulator or a UPS board.
- Never wire 18650 cells or an unregulated supply to the PCB.

> **Never power the node without the LoRa antenna attached. Connect one USB cable at a time:** the ESP32 and the radio share the power rails.

Full guide

For unboxing through deployment, antennas, all detectors, the vibration sensor, mesh commands and Command Center, see the Operator's Guide: github.com/lukeswitz/AntiHunter/blob/main/docs/AntiHunter-Operators-Guide.pdf

- **Firmware, docs, issues:** github.com/lukeswitz/AntiHunter
- **Command Center:** github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO